# AMOD-C power and data conversion

Jason Gallicchio and Claude, rev A, 2026-10-25 github.com/gallicchio/AMOD-and-RFMOD

FOOT FOOT FOOT FOOT

c11 555 VCO
OUT 4.9 kHz
CV +-1 V
moves it

c5 SC integrator
phases IO4, IO8
R = 1 / f C
C = 1 nF
IN +-1 V

c4 SC filter
CLK on IO3
fc = FCLK / 100
to 15 kHz
IN +-1 V

0 to -60 dB
1k 100k

github.com/gallicchio/AMOD-and-RFMOD
handbook and generator

c1 Power
USB-C 5 V only
+5 V 1.5 A
-5 V 40 mA
+3.3 V 300 mA

c6 Delta-sigma
IN +-1.5 V max
CLK IO1, BIT IO7
needs gateware
INT1, INT2 / 4

c10 Cascode
jumper on: CE
off: cascode
IN +-50 mV

c3 Charge pump
IO6 clocks it
OUT about -5 V
R = 1 / f C

c2 Regulators
buck 3.6 V 1.5 MHz
LDO 3.3 V
LOAD 100 mA/V,
1 V max
TRIM +-2 V

c8 Comparator
BIT on IO2
high: IN > DAC
SAR in gateware

c7 Track/hold
IO5 high: track
IO5 low: hold
IN +-1 V
droop 30 mV/s

c12 Rectifier
OUT = |IN|
DC to 100 kHz
IN +-1 V

c9 Differential pair
IN+, IN- +-0.2 V
linear +-25 mV
OUT +-1 V

+1 to -1 V
-0.2V +0.2V

AMOD-C: power and data conversion, switched capacitors and transistors. USB-C 5 V on c1.

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